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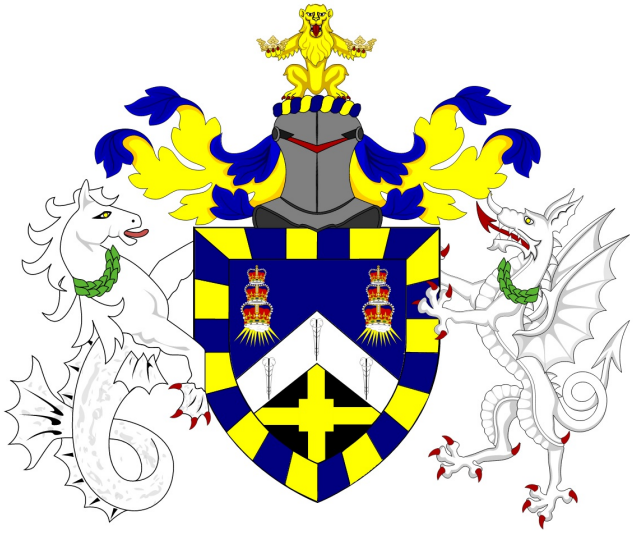
Premier League Match and Season Prediction

A Comparison of Machine Learning, Standard Poisson and Bayesian Poisson Models

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School of Mathematical Sciences

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Declaration of original work

This declaration is made on August 30, 2026.

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Abstract

Predicting football results before a season begins is difficult because team strength can change and individual matches contain considerable uncertainty. However, preseason forecasting can show whether previous season information is sufficient to estimate match outcomes and final league standings. This dissertation examined whether preseason team strength information could predict Premier League match results, goals and final league positions.

The dataset contained 6,460 matches from the 2009/10 to 2025/26 season. Sixteen predictors were created from previous season performance, preseason Elo ratings and promoted team information. Following validation, logistic regression was selected over gradient boosting as the final machine learning model and compared with standard and Bayesian Poisson models. The complete 2025/26 season was reserved for final testing and each model simulated the season 10,000 times.

The standard and Bayesian Poisson models achieved a match result accuracy of 47.4% compared with 46.8% for logistic regression. The Bayesian model achieved the best macro F1-score the lowest log loss and the strongest overall league table performance. It also provided credible intervals describing uncertainty in the expected goal estimates. Although its predictive improvement was small, the Bayesian model provided the most complete analysis. The results also showed that unexpected changes in team performance remained difficult to predict using fixed preseason information.

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1 Introduction

1.1 Background

Football match prediction is a challenging statistical problem because match outcomes are influenced by several interacting factors. Differences in team strength, home advantage, previous performance and attacking and defensive ability can affect the result, while unpredictable events during a match introduce additional uncertainty. A useful prediction model must therefore identify patterns in historical data while recognising that the outcome of an individual match cannot be determined with certainty.

Premier League prediction can be considered at two related levels. The first is match level prediction, where the objective is to estimate whether a match will result in a home win, draw or away win. The second is season level prediction, where match probabilities are used to estimate final league positions and the likelihood of events such as winning the title, finishing in the top four or being relegated. Predicting an entire season is more complex because uncertainty from each of the 380 matches accumulates across the final table.

Different statistical approaches provide different ways of addressing this problem. Machine learning classification models can learn the relationship between team characteristics and the three possible match results. Poisson models instead predict the number of goals scored by each team and convert the resulting expected goal rates into probabilities for individual scorelines and match outcomes. A Bayesian Poisson model extends this approach by representing model parameters and predictions as probability distributions, allowing uncertainty to be incorporated directly into the analysis.

This dissertation compares these approaches within a consistent modelling framework. Historical Premier League results from the 2009/10 season onwards are used to examine match patterns and construct preseason measures of team strength. The engineered predictors include previous season points, goals, shots on target, league position, promoted team status and Elo ratings. The 2025/26 season is kept separate as the final test season so that the models are evaluated on matches that were not used during model training or selection.

1.2 Problem Statement

The main problem investigated in this dissertation is whether historical team level information can be used to predict Premier League matches and reproduce the final structure of a league season. A model may perform reasonably well when predicting individual match outcomes but still produce an inaccurate final league table. Similarly, two models may have identical classification accuracy while producing different probability estimates, goal predictions and season simulations.

It is therefore not sufficient to assess the model using only one performance measure. This project evaluates match result prediction, goal prediction and final league table prediction separately. It also examines whether the additional uncertainty provided by a Bayesian model offers useful information beyond the predictions produced by standard machine learning and Poisson models.

1.3 Aim

The aim of this dissertation is to develop and compare machine learning models - logistic regression and gradient boosting, standard Poisson and Bayesian Poisson models for predicting Premier League match outcomes and simulating the final 2025/26 league table using preseason team strength information.

1.4 Research Questions

The primary research question is: **How effectively can machine learning, standard Poisson and Bayesian Poisson models use preseason team strength information to predict Premier League match outcomes and simulate the final 2025/26 league table?**
The following are supporting questions:

1. Which modelling approach produces the strongest match result prediction?
2. How do the standard Poisson and Bayesian Poisson models compare when predicting home and away goals?
3. How accurately does each model reproduce the final league positions and points?
4. What additional information is provided by Bayesian posterior distributions and credible intervals?
5. Which teams were assigned the highest title, top four and relegation probabilities by each model?

1.5 Objectives

The objectives of the dissertation are:

1. To construct preseason predictors based on previous season performance, promoted-team status and Elo ratings.
2. To establish a majority class baseline and compare logistic regression and gradient boosting using validation data.
3. To develop standard and Bayesian Poisson models for predicting goals and match result probabilities, including Bayesian posterior distributions and credible intervals.
4. To evaluate the models using appropriate prediction measures and simulate the complete 2025/26 season 10,000 times using each final model.

1.6 Report Structure

Chapter 2 - Literature Review

This chapter focuses on previous research on football prediction. It considers machine learning classifiers, Poisson score models, Bayesian approaches and Elo ratings. It also identifies the gap between individual match prediction and complete season simulation.

Chapter 3 - Data and Exploratory Analysis

This chapter describes the Premier League dataset and the data preparation process. It presents the exploratory analysis of match outcomes, goals, home advantage, scorelines and others. It also identifies the pattern that informed the construction of the predictive features.

Chapter 4 - Methodology

This chapter describes the creation of team season summaries, previous season features and preseason Elo ratings. It then explains the chronological training, validation and test split, the baseline and machine learning models, the standard and Bayesian Poisson models, the conversion of expected goals into match result probabilities and the 10,000 season simulations.

Chapter 5 - Results

This chapter compares match results and goal prediction performance, reports the Bayesian convergence diagnostics and posterior predictive checks and presents the simulated league tables and season outcome probabilities.

Chapter 6 - Discussion and Limitations

This chapter interprets the findings and discusses the differences between modelling approaches. It considers the similarity between the standard and Bayesian Poisson predictions, the value of representing uncertainty and the reasons for differences between predicted and actual league tables. The limitations of the data and methodology are examined.

Chapter 7 - Conclusion

This chapter summarises the main findings, assesses whether the research aim and objectives were achieved and proposes possible extensions for future work.

2 Literature Review

Football prediction has been studied using statistical goal models, machine learning classifiers, rating systems and Bayesian methods. These approaches differ in what they predict and how they represent uncertainty. Classification models directly estimate the probability of a home win, draw or away win, whereas Poisson models first estimate the number of goals scored by each team. Bayesian methods additionally represent model parameters as probability distributions while Elo ratings provide a continuously updated measure of relative team strength.

This chapter reviews six studies selected for their direct relevance to the models and predictors used in this dissertation. This covers machine learning, standard Poisson models, Bayesian football models and Elo ratings. It then identifies the research gap addressed by comparing these approaches under the same test conditions and evaluating their predictions at both match and season levels.

2.1 Machine Learning for Premier League Prediction

Machine learning models can identify relationships between match outcomes and predictors such as team form, attacking performance, defensive performance and home advantage. Unlike traditional statistical models, some machine learning algorithms can capture non-linear patterns and interactions without requiring these relationships to be specified in advance. However, their performance depends heavily on the chosen features, training data and validation procedure.

Baboota and Kaur (2019) [1] investigated machine learning methods for predicting Premier League match results. Their study applied feature engineering and exploratory analysis before comparing several classification approaches, including support vector machines, random forests and gradient boosting. The three models each generated possible match outcomes and were evaluated using the ranked probability score, which considers the complete probability distribution rather than only whether the most likely class was correct.

Gradient boosting produced the strongest machine learning performance in their study, with a ranked probability score of approximately 0.216. The corresponding bookmaker score was approximately 0.201, where a lower value represents better probabilistic performance. Consequently, although gradient boosting performed well relative to other machine learning methods the bookmaker probabilities remained difficult to outperform.

Gradient boosting is attractive for football prediction because it can model non-linear relationships and interactions between predictors. For example, the influence of league position may depend on whether a team is playing at home or whether its opponent has recently been promoted. However, the flexibility of gradient boosting also creates a risk of overfitting, particularly when the number of available seasons is limited. Its predicted relationships are also generally less interpretable than those produced by a logistic regression model.

The findings of Baboota and Kaur (2019) [1] motivated the original decision to include gradient boosting as a candidate model in this dissertation. Nevertheless the strongest algorithm is not always guaranteed to be the best when data, features and prediction settings change. Therefore logistic regression ended up being the final machine learning model selected based on the performance of the validation data.

The important difference between the present project and Baboota and Kaur is the construction of predictors. The dissertation concentrates on information that would be available before the beginning of the

predicted season. These variables include previous season performance and preseason Elo ratings. This creates a more restrictive prediction problem because the models are not continually updated with current season results, but it also prevents future matches from influencing the features used for earlier predictions.

2.2 Poisson Models for Football Scores

Poisson models are widely used in football prediction because goals are non-negative count outcomes. Instead of directly predicting whether a match outcome is a home win, away win or a draw, a Poisson model estimates the expected goal scored by each team. The resulting distribution can then be combined to calculate scorelines and match result probabilities.

Maher (1982) [2] provided one of the foundational applications of Poisson modelling to football scores. In this framework, a team's expected number of goals depends on its attacking strength, the opposing team's defensive strength and home advantage. Home and away goals are initially treated as Poisson-distributed variables. The model therefore provides a clear relationship between the underlying strength of the teams and the expected score.

One advantage of Maher's approach is its interpretability. Stronger attacking teams are expected to score more goals while stronger defensive teams reduce their opponents' scoring rates. The model also produces probabilities for exact scorelines from which home win, away win or draw probabilities can be obtained. However home goals and away goals as independent may not fully represent the relationship between the two teams' performance during a match.

Dixon and Coles (1997) [3] developed the Poisson approach further by addressing two important limitations. First, they introduced an adjustment for the dependence between low scoring outcomes particularly scorelines such as 0-0, 1-0, 0-1 and 1-1. These outcomes may not be represented accurately when home and away goals are assumed to be completely independent. Second they allowed recent matches to receive greater weight, recognising that team strength changes over time. Their research demonstrated that apparently small differences in scoreline probabilities can be important when the model is used for betting or probabilistic forecasting. However the additional low score correction and time weighting procedure increases the model's complexity and introduces further parameters that must be estimated.

Koopman and Lit (2015) [4] extended football score modelling through a dynamic bivariate Poisson model. Their approach allowed the scoring of the two teams to be related and permitted team strength to change over time. This is useful because football teams are not static: changes in players, managers and tactics can affect attacking and defensive performance during a season. The dynamic bivariate approach is more flexible than a fixed independent Poisson model but it is also more computationally demanding. It requires additional assumptions about how team strengths evolve and more complex estimation procedures. This may improve forecasts when recent match information is available but the model is less straightforward to implement and interpret.

The standard Poisson model developed in this dissertation follows the general principle established by Maher (1982): expected home and away goals are estimated before being converted into score and result probabilities. However it uses separate Poisson regression models with preseason team strength predictors rather than estimating a separate attack and defence parameter for every club. The same predictors were used for the home goal and away goal models allowing the estimated effect to be compared.

The model does not apply the Dixon-Coles low score adjustment and does not use a dynamic bivariate

structure. Home and away goal counts are therefore treated as conditionally independent after accounting for the predictors. This is a deliberate simplification that supports interpretability and provides a consistent comparison with the Bayesian model.

2.3 Bayesian Models for Football Prediction

Bayesian modelling offers an alternative way of estimating football models and representing uncertainty. Instead of producing only a single estimated value for each coefficient, a Bayesian model combines prior distributions with observed data to obtain posterior distributions. These distributions describe the range of parameter values that remain plausible after the data have been observed.

Baio and Blangiardo (2010) [5] developed a Bayesian hierarchical model for predicting football results. Their model represented each team through attacking and defensive strength parameters, which were used to determine the expected home and away goal rates. Because the model was hierarchical the team parameters were related through common population distributions. This produces partial pooling, shrinking team estimates for teams with less data towards the league average.

Partial pooling can be valuable in football prediction because the amount of information available for each club is not always equal. For example newly promoted teams may have limited recent data from the league being predicted. Hierarchical estimation can reduce extreme estimates based on small samples. However, excessive shrinkage may make teams appear more similar than they actually are. Baio and Blangiardo therefore also considered a more flexible mixture structure to represent differences between teams.

A major benefit of their Bayesian approach is that uncertainty is retained throughout the prediction process. Posterior distributions can be used to produce credible intervals for expected goals and other model quantities. Prediction can also incorporate parameter uncertainty instead of relying entirely on one fitted set of coefficients.

The Bayesian model in this dissertation is influenced by this principle but is not an exact reproduction of the hierarchical attack and defence model used by Baio and Blangiardo (2010) [5]. The present model is a Bayesian Poisson regression in which the expected home and away goal rates depend on the same standardised preseason predictors used by the standard Poisson model. Prior distributions are assigned to the intercepts and the feature coefficients and posterior sampling is used to estimate their distribution.

Using common predictors makes the standard and Bayesian Poisson models directly comparable. Differences in their results are less likely to be caused by completely different input information. Posterior samples can also be carried forward into the match and season predictions, allowing uncertainty in the coefficients and expected goal rates to influence the simulation results.

Bayesian estimation does not automatically guarantee more accurate predictions. Its performance continues to depend on the suitability of the predictors, prior distribution and Poisson likelihood. If the goal model is misspecified representing its parameters probabilistically does not remove the underlying limitation. Sampling diagnostics and predictive performance must therefore be examined alongside the credible intervals.

2.4 Elo ratings as a Measure of Team Strength

Elo is a rating system that assigns each team a numerical measure of its relative strength. Ratings are updated after every match according to the result and the difference between the teams' pre-match ratings. A team gains rating points after performing better than expected and loses points after performing worse than expected. Unexpected results produce larger changes than results that were already likely. In football applications, home advantage can also be added when calculating the expected results.

Hvattum and Arntzen (2010) [6] examined the use of Elo ratings for predicting association football match results. They demonstrated that Elo ratings could be used as explanatory information in an ordered logistic regression model to generate probabilities for home win, away win or draw. Their research supports the use of Elo as a compact measure of relative strength rather than only as a system for ranking teams.

An important advantage of Elo ratings is that defeating a strong team provides a larger increase in rating than defeating a weaker team. Similarly losing to a stronger team provides a smaller decrease than to a weaker team. It gives you more information than just looking at past wins and home advantage can be included in calculating the expected outcome. However Elo compresses several aspects of performance into a singular value. It does not differentiate between attacking and defensive strength. Its values also depend on choices such as initial rating, home advantage, update factor and how a new team entering the league is treated.

In this dissertation, preseason Elo is used alongside previous season points per game, goals and other features. Elo is therefore treated as a complementary feature rather than the sole measure of team quality. The ratings are calculated using earlier match results and are fixed at their preseason values for the season that is being predicted. This prevents results from the test season leaking into its match predictors.

2.5 Comparison of the Literature and Research Contribution

Study	**Method**	**Relevance to this dissertation**
Baboota and Kaur (2019)	Machine-learning classification	Motivates the original gradient boosting while also showing the need for data specific validation.
Maher (1982)	Poisson goal model	Provides the foundation for goal based prediction but relies on restrictive scoring assumptions.
Dixon and Coles (1997)	Adjusted Poisson model	Improves the traditional Poisson structure but introduces additional complexity.
Koopman and Lit (2015)	Dynamic bivariate Poisson model	More flexible than independent Poisson regression but more computationally demanding.
Baio and Blangiardo (2010)	Bayesian hierarchical Poisson model	Supports the use of Bayesian modelling to represent uncertainty.
Hvattum and Arntzen (2010)	Elo ratings and ordered logistic regression	Supports the inclusion of preseason Elo as a team-strength predictor.

Table 2.1: Summary of the selected football-prediction literature

Table 2.1 summarises the six studies and their relation to this dissertation.

The reviewed literature shows that each approach offers different advantages. Machine learning models can

capture complex relationships but may overfit and can be difficult to interpret. Standard Poisson models show a connection between expected goals, scorelines and match outcomes, although their distributional assumption is restrictive. Bayesian models quantify parameter uncertainty but are dependent on the suitability of the likelihood and priors. Elo ratings offer a compact measure of relative strength but cannot represent the attacking and defensive strength separately.

The studies use different datasets, time period, predictors and evaluation measures, therefore the performances cannot be compared directly. A model that seems superior in one study might simply have had access to better data or faced an easier prediction task.

Within the six studies, none performs the combined comparison of the three models in this dissertation to predict a complete Premier League season. The literature concentrates mainly on individual match outcomes, score distributions or betting forecasts. Less attention is given to whether match probabilities accumulate into an accurate final league table.

This dissertation addresses that gap by comparing the three modelling approaches under a consistent framework. All final models use preseason information and are evaluated on the same 2025/26 season. Their performance is examined at three levels: match results prediction, goal prediction and final league table prediction. The probabilities produced by each model are subsequently used in 10,000 season simulations, where a detailed procedure is presented in Chapter 4.

Instead of copying the complex past models, this dissertation tests whether simpler, interpretable models supplied with the same preseason data produce different match forecasts, league tables and uncertainty estimates.

3 Data and Exploratory Data Analysis

This chapter describes the Premier League dataset used in the dissertation and presents the exploratory analysis completed before model development. The analysis examines the distribution of match results, home advantage, goal scoring patterns, shots on target, team performance and the results achieved by promoted clubs. The 2025/26 test season was excluded from the exploratory analysis to prevent information from the final evaluation period from affecting the modelling decisions.

3.1 Data Source, Variables and Preparation

Historical Premier League match data were obtained from Football-Data [7]. The data were provided as separate CSV files for each season with one row representing one completed match. The files contained information about the participating teams, full-time goals, match results and shots on target.
The complete dataset covered the 17 Premier League seasons from 2009/10 to 2025/26 and contained 6,460 matches. The 2025/26 season was separated before exploratory analysis and model development because it was reserved for the final out of sample testing. The remaining 16 seasons contained 6,080 matches with 380 matches each season.

Dataset	Seasons	Number of seasons	Number of matches
Complete dataset	2009/10–2025/26	17	6,460
Exploratory dataset	2009/10–2024/25	16	6,080
Final test season	2025/26	1	380

Table 3.1: Coverage of the Premier League dataset

Only variables relevant to the exploratory analysis, feature construction and prediction models were retained. The variables are presented in Table 3.2

Variable	Description	Use in the project
`Season`	Premier League season in which the match occurred	Chronological division and feature construction.
`Div`	League division code, recorded as `E0`	Confirms that the observation is a Premier League match.
`Date`	Date on which the match was played	Chronological ordering and Elo calculation.
`HomeTeam`	Name of the home team	Team identification and home feature assignment.
`AwayTeam`	Name of the away team	Team identification and away feature assignment.
`FTHG`	Full-time goals scored by the home team	Target for the home goals models.
`FTAG`	Full-time goals scored by the away team	Target for the away goals models.
`FTR`	Full-time result: home win, draw or away win	Target for the classification models.
`HST`	Home team shots on target	Construction of previous season team statistics.
`AST`	Away team shots on target	Construction of previous season team statistics.

Table 3.2: Variables retained from the original match data

Several preparation steps were completed before the data were analysed. Dates in the earlier season files used different formats, so they were converted into a consistent day-first format and sorted as dates. A season label was added to every file before the selected variables were retained and the individual datasets were combined. The important fields were checked for missing values. No missing values were found in the match date, team names, goals, results or shots on target variables. The variable types and final dataset dimensions were also examined and each season was confirmed to contain 380 matches.
Finally the 2025/26 matches were removed from the exploratory and development dataset and stored separately as the final test season. This chronological separation ensured that results from the test season did not influence the exploratory findings, feature construction, model training or model selection.

3.2 Match Outcomes, Goals and Shots on Target

The distribution of match outcomes was examined using the 6,080 matches from 2009/10 to 2024/25 as shown in Figure 3.1, home wins were the most frequent outcome occurring for a total of 2,751 matches. Away wins occurred 1,864 times while 1,465 ended as draws. These values equate to 45.25% home wins, 30.66% away wins and 24.10% draws. This provides evidence that an overall home advantage exists as there is a difference of 14.59% of the home team winning over the away team. However, there is a also the fact less than half the matches resulted in home wins which demonstrates that there's uncertainty to the outcomes.

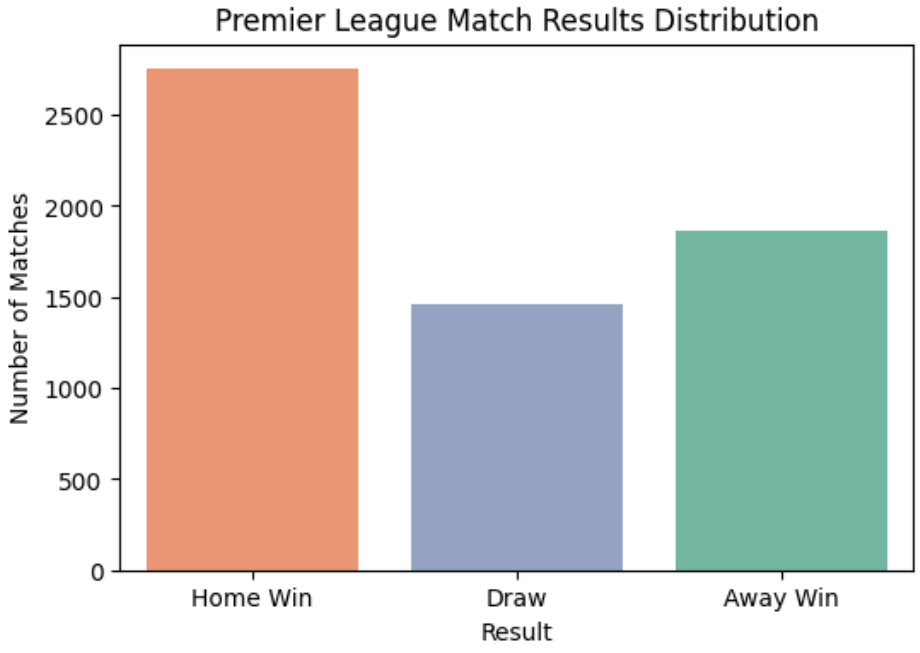


Figure 3.1: Distribution of Premier League match results, 2009/10–2024/25

The percentage of each outcome was also calculated separately for each season. Figure 3.2 shows that home wins were more frequent in 15 of the 16 seasons. The highest home win percentage was 50.79% in 2009/10 while the lowest was 37.89% in 2020/21.
The 2020/21 season was the only season in which away wins were more frequent than home wins, with the

away win at 40.26%. Home wins then increased to 48.42% in 2022/23 before dropping to 40.79% in 2024/25. These variations demonstrate that the size of home advantage was not constant across the historical period.

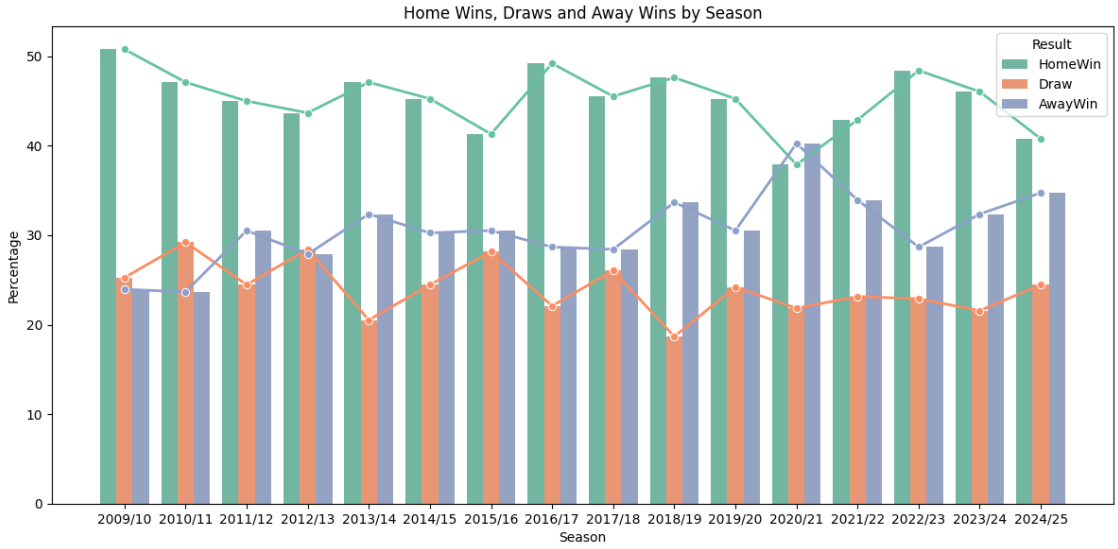


Figure 3.2: Percentage of home wins, draws and away wins by Premier League season

Draws were the least frequent outcome overall, although their proportion did vary each season. The class imbalance between home wins, draws and away wins is important for model evaluation. A model could obtain a misleadingly reasonable accuracy by favouring the most common home win category while performing poorly when identifying draws. For this reason, later model comparisons consider macro F1-score and log loss in addition to classification accuracy. The observed difference between home and away results supports the inclusion of separate home team and away team predictors. It also justifies estimating home and away goals separately in the Poisson models rather than assuming that the two teams have identical scoring conditions.

The distribution of match outcomes was further examined by considering the number of goals scored. Across the 6,080 matches in the exploratory dataset a total of 17,026 goals were scored. This produced an overall average of approximately 2.80 goals per match, however the scoring rate was not identical in every season. Most seasons recorded an average between 2.60 and 2.90 goals per match. The 2023/24 season produced the highest average in the period with 3.28, whereas 2014/15 had the lowest with 2.57. This variation indicates that the Premier League scoring environment can change between seasons.

The home and away goal averages were also compared. As shown in Figure 3.3 home teams scored an average of approximately 1.56 while away teams scored 1.24. The difference provides further evidence for home advantage.

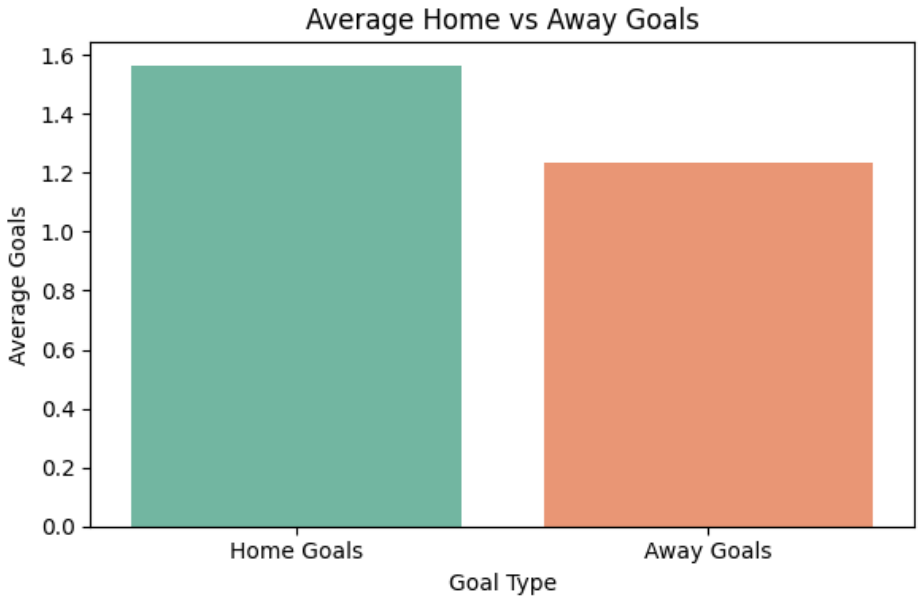


Figure 3.3: Average home and away goals per Premier League match

Goal totals also differed according to the match result. Matches won by the home team contained an average of 3.06 while an away victory contained 2.98 and draws contained 2.09 indicating draws were generally lower scoring matches.

The most common scoreline was 1-1, which occurred in a total of 650 matches and was roughly 10.69% of the exploratory data. This was followed by a 1-0 with 577 matches and 2-1 with 511 matches. Figure 3.4 shows low scoring matches accounted for most of the Premier League matches.

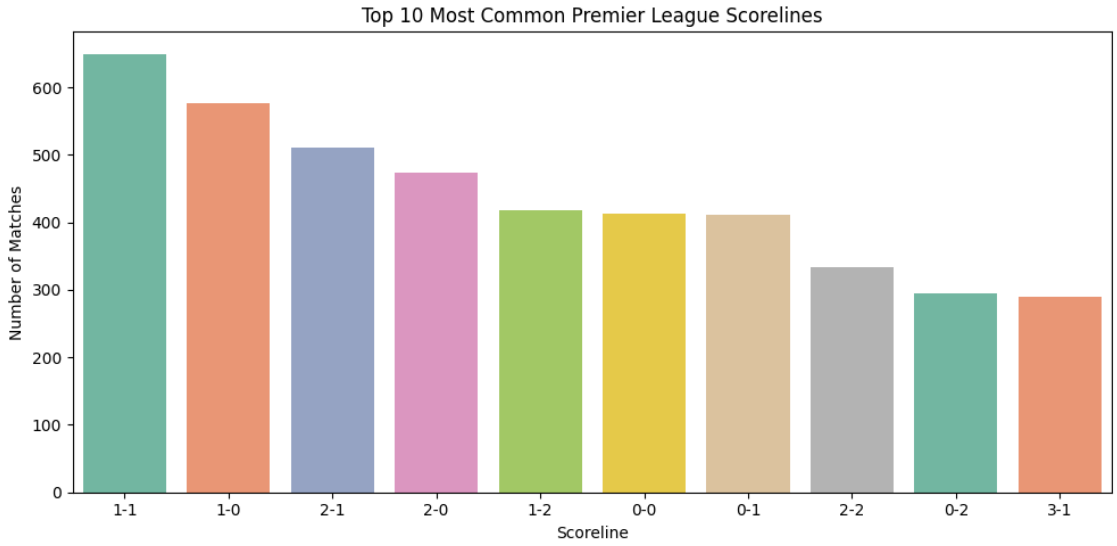


Figure 3.4: Ten most common Premier League scorelines, 2009/10–2024/25

The concentration of results around relatively small goal counts supports the use of Poisson models for home and away goals. It also demonstrates why the probabilities assigned to low scoring outcomes are important when converting expected goals into home win, draw and away win probabilities.
Since goals are a result of attacking opportunities, the relationship between shots on target and goals was then inspected. Shots on target were examined to determine whether they provided useful information about attacking performance. The correlation for home goals to home shots on target was 0.485 and for the away was 0.490. They both have a moderately positive relationship but neither is sufficiently strong to suggest that shots on target alone could determine the final score. Goalkeeper performance, finishing quality of the attackers and other events can also affect how many shots end up as goals.
Shots on target were also grouped according to full-time results. As shown in Figure 3.5, home teams averaged 6.77 shots on target when they won compared to the 3.67 for the away team. In an away win the away team averaged 5.87 while the home team averaged 4.19. Drawn matches were closer with the home team at 5.07 and the away team at 4.42 shots on target.

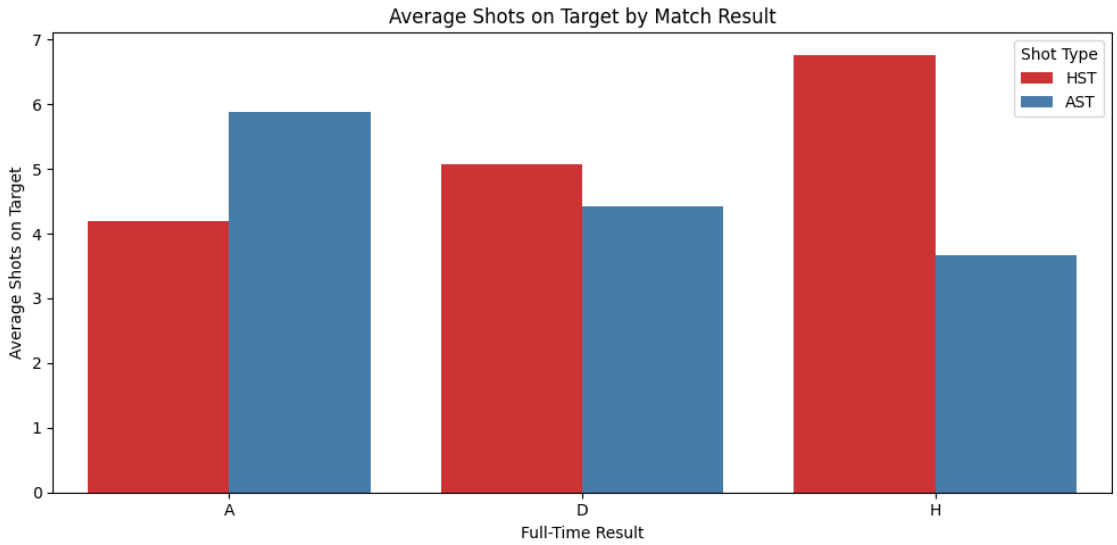


Figure 3.5: Average home shots on target (HST) and away shots on target (AST) by match result

These results demonstrate that the winning team generally produced more shots on target than its opponent. Nevertheless considerable variation remained due to matches with similar shot totals could produce different scores and results. Shots on target can therefore be treated as an informative measure of attacking performance rather than a direct measurement of future goals.

3.3 Team Performance and Promoted Club

Team performance was examined by combining home and away goals for every club appearing between 2009/10 and 2024/25. Figure 3.6 shows that Manchester City scored the most with 1,375. Followed by Liverpool with 1,199 then Arsenal, Chelsea and Tottenham with 1,149, 1,107 and 1,069 respectively. Manchester City also recorded the largest goal difference at +825 followed by Liverpool at +557. These results show the attacking and defensive strength between Premier League clubs.

The cumulative totals should be interpreted carefully as not every team has participated in the Premier League for the same number of seasons. Teams that were continuously present throughout the period had more opportunities to score and concede goals, than teams that had one or two seasons. The figure is more of a description of performance rather than a direct comparison of scoring efficiency.

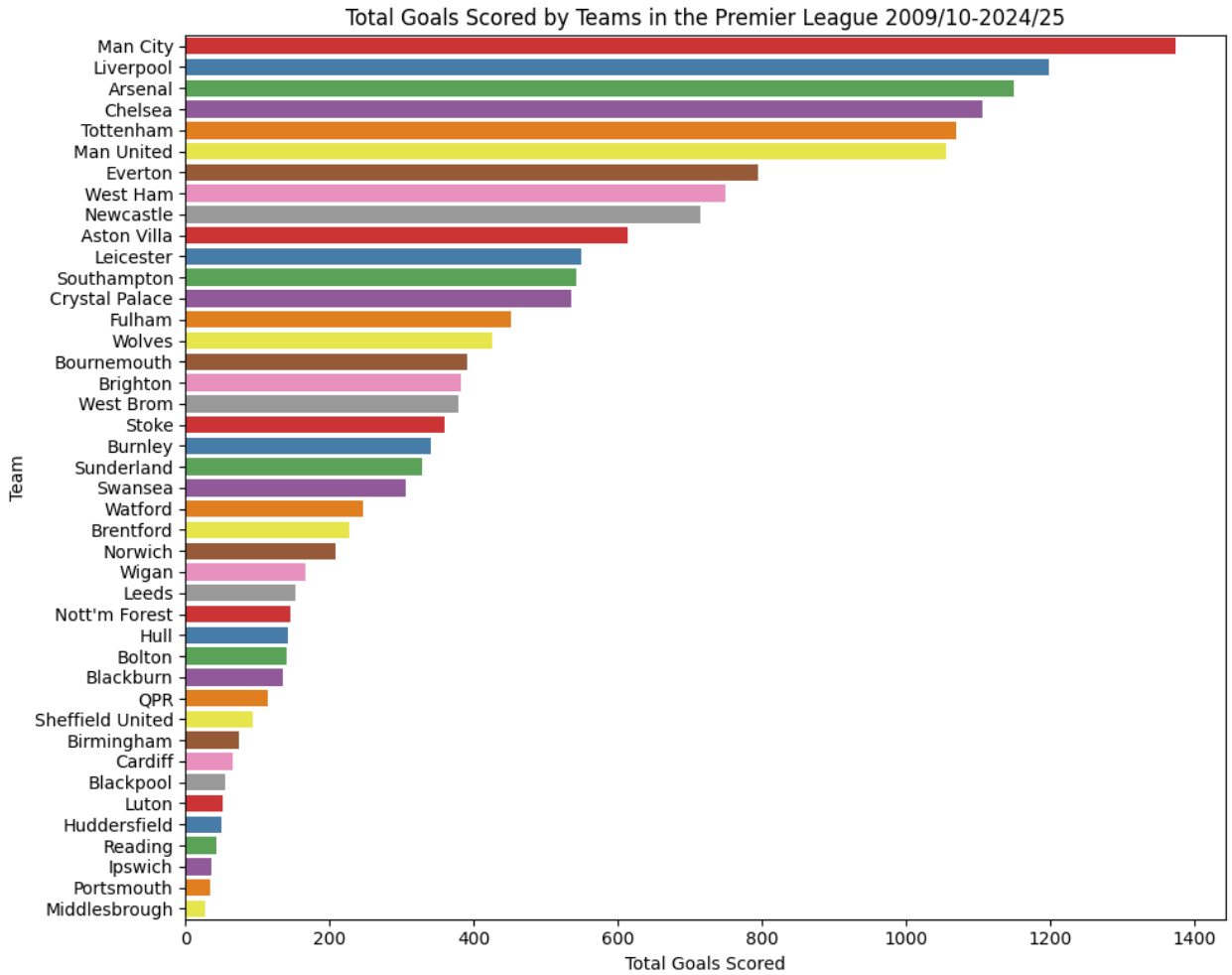


Figure 3.6: Total goals scored by Premier League teams, 2009/10–2024/25

Promoted clubs were analysed separately because they generally entered the Premier League without previous season performance from the same competition. Across the 16 exploratory seasons there were 48 promoted team entries, representing 29 unique clubs. A club was classified as having survived if it finished 17th or above in the league.

As shown in Figure 3.7 26 promoted clubs survived while 22 were relegated immediately. This produced a survival rate of 54.17%, promoted status therefore represented a substantial disadvantage although promotion did not make relegation inevitable.

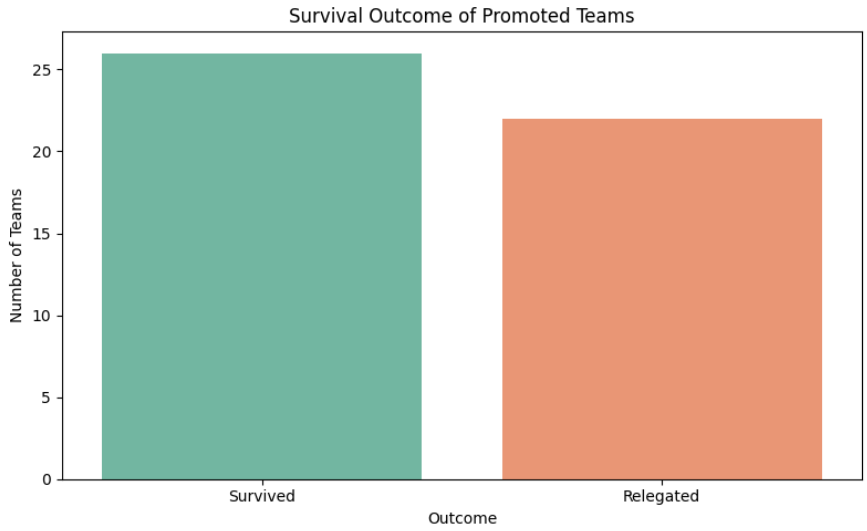


Figure 3.7: Survival outcomes of promoted Premier League teams, 2009/10–2024/25

Table 3.3 compares the average performance of all promoted clubs and separated them into those that survived and those that were relegated.

Outcome	**Average points**	**Average position**	**Average goal difference**
All promoted clubs	36.27	15.67	−25.54
Survived	44.23	12.85	−13.23
Relegated	26.86	19.00	−40.09

Table 3.3: Average performance of promoted Premier League clubs

Promoted clubs that survived obtained approximately 17.37 more points on average than those that were relegated. The largest distinction was defensive where teams that survived conceded 56.46 goals and relegated teams conceded 75.59 goals.

4 Methodology

This chapter describes the models used to construct the preseason predictors, develop the competing prediction models and simulate the 2025/26 Premier League season. The modelling process consisted of five main stages: feature engineering and data preparation, machine learning model selection, standard Poisson, Bayesian Poisson modelling and final evaluation and season simulation. All predictors were constructed using information available before the beginning of the season being predicted. This chronological approach prevented results from a target season from influencing its own match predictions.

4.1 Preseason Feature Engineering and Data Preparation

The cleaned match level dataset described in Chapter 3 contained one row for each Premier League match. However, the models required variables representing the team strength before the beginning of the season, team season summaries were then constructed before these statistics were attached to the following season's matches. For each club and season, home and away performance were combined to calculate matches played, points, goals scored, goals conceded, shots on target for and shots on target conceded. Three points were awarded for a win, one for a draw and zero for a loss. The seasonal totals were converted into per game measures using

$$x_{i,s}^{PG} = \frac{x_{i,s}}{M_{i,s}},$$

where $x_{i,s}$ is the seasonal total for team i in season s and $M_{i,s}$ is the number of matches played by that team. This produced points, goals scored, goals conceded, shots on target for and shots on target conceded per game for all five performance variables.

The final league position of each team was also calculated. Teams were ordered first by points followed by goal difference and goals scored. Goal difference was used to determine the historical league position but was not included as a separate predictor in the final models.

For every match in a target season the previous season statistics of the home and away teams were added to the match record. For example, predictors for a match in 2024/25 were constructed using the team performance from 2023/24. These values were fixed throughout the target season. Consequently later matches in a season did not benefit from information obtained from earlier matches in the season.

Promoted clubs required different treatment because they did not have statistics from the immediately preceding Premier League season. Each promoted club was assigned the average performance of the three teams that finished at the bottom of the previous Premier League season. This produced a conservative estimate of likely Premier League strength while maintaining the same set of predictors for every team. A binary promoted team variable was also created, taking the value one for a promoted club and zero otherwise. As discussed in Section 2.4, the treatment of teams entering the league is an important part of measuring team strength. The promoted team indicator therefore allowed the models to distinguish these clubs from established Premier League teams event when their assigned performance measure were similar..

In addition to previous season performance, preseason Elo ratings were created as a continuously updated measure of team strength. Elo ratings adjust after each result according to the strength of the opponent and whether the result was expected. Their use in football prediction is supported by Hvattum and Arntzen (2010) [6], who demonstrated that Elo ratings could provide useful information for probabilistic match

prediction. Each established team began with an Elo rating of 1500, while the promoted teams in the first season began 50 points lower. Before each match, the expected home team result E_H was calculated as

$$E_H = \frac{1}{1 + 10^{\frac{R_A - (R_H + 80)}{400}}}$$

where R_H and R_A represent the rating of the home and away teams. An 80-point adjustment was added to the home team's rating to represent the home advantage. The actual home team result S_H was assigned a value of one for a home win, 0.5 for a draw and zero for an away win. The ratings were then updated using

$$R'_H = R_H + 20(S_H - E_H) \qquad \text{and} \qquad R'_A = R_A - 20(S_H - E_H).$$

where R'_H and R'_A are the new ratings after the match has taken place. The update factor of 20 controlled the size of rating change. An unexpected result therefore produced a larger adjustment than a result the rating system already considered likely.

At the beginning of each new season the ratings of continuing Premier League teams were moved 25% towards the initial rating of 1500

$$0.75R^{\text{end}} + 0.25(1500)$$

This regression towards the mean reduced the influence of results from older seasons and recognised that team strength could change over summer. After this regression was applied the lowest rating among the continuing Premier League teams was identified. A club entering the dataset was assigned a preseason Elo rating of 50 less than this

$$R^{\text{pre}}_{\min,s} - 50$$

A different procedure was used when a previously relegated club returned to the Premier League. Its most recent Premier League Elo rating was blended with the promoted team starting rating. If the club had been absent for $a_{i,s}$ seasons the weight assigned to its previous rating was

$$w_{i,s} = 0.7^{a_{i,s}}$$

Its returning preseason rating was therefore calculated as

$$w_{i,s} R^{\text{last}} i + (1 - w_{i,s}) R^{\text{prom}}_s$$

where R^{last}_i is the club's Elo rating at the end of its most recent Premier League season. The weight placed on this historical rating decreased with every season spent outside the Premier League. The resulting blended rating was used directly and was not capped by the rating of the lowest continuing team. This allowed a returning club to retain evidence of strong previous Premier League performance, even if it had previously been relegated during a comparatively strong season. Clubs returning after a longer absence were still moved progressively closer to the standard promoted team rating.

The calculated preseason Elo rating was fixed for every match in the target season. Historical results during that season were used to update the internal Elo ratings required for constructing the following season's preseason values, but they did not alter the predictors already assigned to matches in the current season.

The final modelling dataset contained 16 predictors for both home and away teams, these were previous season points per game, goals scored per game, goals conceded per game, shots on target for per game, shots on target conceded per game, final league position, preseason Elo rating and promoted team status.
Because previous season information was required, the first modelled season was 2010/11 with the 2009/10 season providing the initial historical statistics. The data were divided chronologically as follows:

- The 2010/11 to 2023/24 seasons formed the training dataset, containing 5,320 matches.
- The 2024/25 season formed the validation dataset, containing 380 matches.
- The 2025/26 season formed the final test dataset, containing 380 matches.

The validation season was used for model comparison and machine learning model selection. After the final model specifications had been selected the training and validation data were combined, producing 5,700 matches for final model estimation. The 2025/26 test season remained separate until final evaluation. This ensured that the final reported performance represented genuinely out of sample prediction.

4.2 Baseline and Machine Learning Models

The match result target contained three categories: home win, draw and away win. A majority class, logistic regression and gradient boosting were initially evaluated. Each approach produced a predicted result and probabilities for three possible outcomes.

4.2.1 Majority Class Baseline

A majority class baseline was constructed to provide a simple benchmark for the predictive models. The baseline identified the most frequent result in the training data and assigned that result to every validation match. Because home wins were the most frequent the baseline always predicted a home win.
For probabilistic evaluation, every validation match was assigned the observed training proportions of home wins, draws and away wins. This allowed the baseline to be assessed using log loss as well as classification accuracy. The baseline was not considered for selection as the final machine learning model, it was included to decide if the other models would have an improvement over a simple prediction rule.

4.2.2 Logistic Regression

Multinomial logistic regression estimates the probability of each possible outcome as a function of the predictors. It extends binary logistic to targets containing more than two categories and produces probabilities that sum to one across the possible classes. (Hastie, Tibshirani and Friedman, 2009) [8].
For match i and result category k, the probability was presented as

$$P(Y_i = k \mid \mathbf{x}_i) = \frac{\exp\left(\beta_{0,k} + \mathbf{x}_i^T \boldsymbol{\beta}_k\right)}{\sum_{j \in \{H,D,A\}} \exp\left(\beta_{0,j} + \mathbf{x}_i^T \boldsymbol{\beta}_j\right)}, \qquad k \in \{H, D, A\}.$$

where $\mathbf{x}_i$ is the vector of 16 preseason predictors, β_{0k} is the intercept for result k which contains H for home win, D for draw and A for away win and $\boldsymbol{\beta_k}$ contains its feature coefficients. The predicted result was the category assigned the highest probability. The predictors were standardised before logistic regression was

fitted because they were measured on different scales. For example Elo ratings were approximately centred around 1500 while promotion status was represented by 0 or 1. Each continuous predictor was transformed using

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}$$

where $\bar{x}_j$ and s_j were the training mean and standard deviation of predictor j. Standardisation and logistic regression were placed in the same processing pipeline so that the scaling values were only learned from the training data. This prevented information from validation or test season influencing the models fitting. The maximum number of iterations was set to 1,000 while the remaining settings used the default implementation values.

4.2.3 Gradient Boosting

Gradient Boosting is an ensemble method that constructs a sequence of decision trees. Each new tree is fitted to reduce the errors remaining from the preceding trees, and their predictions are combined to produce the final result probabilities (Friedman, 2001) [9]. Its application to Premier League match prediction was also supported by Baboota and Kaur (2019) [1].

For result category k the final prediction M boosting stages can be written as:

$$F_k(\mathbf{x}_i) = F_k^{(0)}(\mathbf{x}_i) + \eta \sum_{m=1}^{M} h_{m,k}(\mathbf{x}_i), \qquad k \in \{H, D, A\}.$$

where $F_k^{(0)}$ is the initial score for category k, $h_{m,k}$ is the decision tree added at boosting stage m, η is the learning rate and M is the total number of boosting stages Each new tree was fitted to reduce the multinomial log loss error of the existing ensemble. The final scores were converted into match result probabilities using:

$$P(Y_i = k \mid \mathbf{x}_i) = \frac{\exp\left(F_k^{(M)}(\mathbf{x}_i)\right)}{\sum_{j \in \{H,D,A\}} \exp\left(F_j^{(M)}(\mathbf{x}_i)\right)}, \qquad k \in \{H, D, A\}.$$

The predicted results was the category with the highest probability. Unlike logistic regression the function $h_{m,k}$ were decision trees rather than linear combinations of the predictors. This allowed the model to represent non-linear relationships and interactions. For example the effect of league position could vary according to a team's Elo rating or promoted status.

The gradient boosting classifier was fitted using the 16 preseason predictors. A random seed of 42 was specified to support reproducibility, while the remaining hyper parameters use the default implementation values. Feature importance values were also extracted to examine which predictors contributed most strongly to the fitted model. The importance values were used for interpretation rather than for removing predictors or selecting the final model.

4.2.4 Model Selection

The two candidate models were trained using matches from 2010/11 to 2023/24 and evaluated on the chronologically separate 2024/25 validation season. Accuracy, macro F1-score, log loss and confusion matrices were

examined. Log loss was used as the primary selection measure because the complete outcome probabilities were subsequently required for season simulation. If the models produced equal log loss then the model with the higher validation accuracy was selected.

The majority class baseline was included in the validation comparison but was excluded from final model selection. After the candidate model had been selected, the modelling pipeline was then fitted again using the combined training and validation dataset of 5,700 matches. It was then used to produce home win, draw and away win probabilities for the 380 matches held in 2025/26 test season.

4.3 Standard Poisson Model

A standard Poisson regression model was developed to predict the number of goals scored by each team per match. Poisson models are suitable for football scores because goals are non-negative count outcomes and typically occur at relatively low frequencies. Their use in football prediction was established by Maher (1982) [2], who modelled scoring rates using measures of team strength and home advantage.

Separate models were fitted for home goals and away goals. Let $Y_{H,i}$ and $Y_{A,i}$ represent the number of goals scored by the home and away teams in match i. The model assumed:

$$Y_{H,i} \sim \text{Poisson}(\lambda_{H,i}) \qquad \text{and} \qquad Y_{A,i} \sim \text{Poisson}(\lambda_{A,i})$$

where $\lambda_{H,i}$ and $\lambda_{A,i}$ are the expected home and away goals for match i. For a Poisson distributed variable the probability of observing y goals is:

$$P(Y = y \mid \lambda) = \frac{\exp(-\lambda)\lambda^y}{y!}, \qquad y = 0, 1, 2, \ldots$$

A logarithmic link function connected the expected goal rate to the match predictors:

$$\log(\lambda_{H,i}) = \beta_{0,H} + \mathbf{x}_i^T \boldsymbol{\beta}_H \qquad \log(\lambda_{A,i}) = \beta_{0,A} + \mathbf{x}_i^T \boldsymbol{\beta}_A$$

Equivalently,

$$\lambda_{H,i} = \exp\left(\beta_{0,H} + \mathbf{x}_i^T \boldsymbol{\beta}_H\right) \qquad \lambda_{A,i} = \exp\left(\beta_{0,A} + \mathbf{x}_i^T \boldsymbol{\beta}_A\right)$$

Here, $\mathbf{x}_i$ is the vector containing the 16 match level predictors described in Section 4.1. The vectors $\boldsymbol{\beta}_H$ and $\boldsymbol{\beta}_A$ contain separate coefficients for the home goal and away goal models allowing the same predictor to have different effects on home and away scoring.

The predictors were standardised using a StandardScaler fitted only to the relevant training data. Each goal model was implemented using Poisson regression with an L2 regularisation value of $\alpha = 1$ and maximum of 1,000 optimisation iterations. Regularisation was included to reduce instability caused by correlated predictors and discourage excessively large coefficients.

The home goal model was trained using full-time home goals as its target while the away goal model used full-time away goals. The initial models were fitted to the training partition and used to generate expected home and away goals for the validation season. Their goal predictions were evaluated against a simple mean goal baseline, which predicted every validation match using the average home and away goal totals observed in the training data.

4.3.1 Converting Expected Goals into Match Probabilities

The expected goal rates were converted into probabilities for individual scorelines. Conditional on the match predictors, home and away goals were treated as independent. Consequently, the probability of a home score of h and away score of a was calculated as:

$$P(Y_{H,i} = h), \qquad P(Y_{A,i} = a)$$

For each match probabilities were calculated for home and away goal totals from 0 to 10. The values formed an 11×11 score probability matrix. Because the matrix excluded the very small probabilities associated with either team scoring more than ten goals it was normalised so that its included probabilities summed to one:

$$p_i(h, a) = \frac{P(Y_{H,i} = h)P(Y_{A,i} = a)}{\sum_{r=0}^{10}\sum_{s=0}^{10} P(Y_{H,i} = r)P(Y_{A,i} = s)}, \qquad h, a \in \{0, \ldots, 10\}.$$

The three match result probabilities were then obtained by summing the appropriate cells of the score probability matrix:

$$\begin{aligned} P(\text{Home Win}) &= \sum_{h>a} p_i(h, a), \\ P(\text{Draw}) &= \sum_{h=a} p_i(h, a), \\ P(\text{Away Win}) &= \sum_{h<a} p_i(h, a). \end{aligned}$$

The predicted result was the class with the highest probability among the home win, draw and away win. The most likely score was identified as the scoreline corresponding to the largest individual value in the probability matrix. The model did not include Dixon-Coles low score adjustment or a bivariate dependence term. Home and away goals were therefore conditionally independent after accounting for the preseason predictors. This provided a comparatively transparent model and maintained consistency with the feature based Bayesian Poisson model in the following section.

4.4 Bayesian Poisson Model

A Bayesian Poisson regression model was developed to predict home and away goals while retaining uncertainty in the model parameters. Bayesian football models have previously been used to determine team strength and predictive uncertainty through posterior distribution by Baio and Blangiardo (2010) [5]. However the model used in this dissertation was not a hierarchical team specific model. It used the same feature based regression structure and the same 16 preseason predictors and the standard Poisson model allowing the two approaches to be compared.

4.4.1 Model structure and prior distribution

Let $Y_{H,i}$ and $Y_{A,i}$ represent the home and away goals in match i. The likelihood was specified as:

$$Y_{H,i} \sim \text{Poisson}(\lambda_{H,i}) \qquad Y_{A,i} \sim \text{Poisson}(\lambda_{A,i})$$

The expected goal rates were connected to the standardised predictors through logarithmic link functions:

$$\log(\lambda_{H,i}) = \beta_{0,H} + \mathbf{x}_i^T \boldsymbol{\beta}_H \qquad \log(\lambda_{A,i}) = \beta_{0,A} + \mathbf{x}_i^T \boldsymbol{\beta}_A$$

Therefore,

$$\lambda_{H,i} = \exp\left(\beta_{0,H} + \mathbf{x}_i^T \boldsymbol{\beta}_H\right) \qquad \lambda_{A,i} = \exp\left(\beta_{0,A} + \mathbf{x}_i^T \boldsymbol{\beta}_A\right)$$

Separate intercepts and coefficient vectors were estimated for home and away goals. The prior mean for each intercept was based on the logarithm of the corresponding mean goal total in the training data:

$$\beta_{0,H} \sim \mathcal{N}\left(\log(\bar{y}_H), 0.3^2\right) \qquad \beta_{0,A} \sim \mathcal{N}\left(\log(\bar{y}_A), 0.3^2\right)$$

where $\bar{y}_H$ and $\bar{y}_A$ are the average home and away goals in the relevant training partition. This centred the intercept priors on historically plausible scoring rates without using information from the validation or test targets. The home and away feature coefficients were assigned independent zero-centred normal priors:

$$\beta_{H,m} \sim \mathcal{N}(0, 0.3^2) \qquad \beta_{A,m} \sim \mathcal{N}(0, 0.3^2) \qquad m = 1, \ldots, 16$$

These priors regularised coefficients towards zero while allowing the observed data to determine whether each predictor increased or decreased the expected goal rate. Standardising the predictors before estimation ensured that the same prior scale could be applied meaningfully to all 16 coefficients.

4.4.2 Posterior estimation and diagnostic checks

The Bayesian model was implemented in PyMC and estimated using the No-U-Turn Sampler, an adaptive form of Hamiltonian Monte Carlo [10], four sampling chains were used. Each chain contained 1,000 tuning iterations followed by 1,000 retained posterior draws producing 4,000 posterior samples. Sampling using two processor cores a target acceptance rate of 0.9 and a random seed of 42 to support reproducibility.

Convergence was assessed using the potential scale reduction statistic $\hat{R}$ the bulk and tail effective sample sizes and the number of divergent transitions. Values of $\hat{R}$ close to one indicate that the separate chains produced similar posterior distributions. Effective sample size measures how much independent information is contained in the correlated simulation draws while the divergent transitions can indicate difficulty exploring the posterior distributions.

Posterior predictive checks were also conducted on the training data. Goal totals were generated from the fitted posterior distributions and compared with the observed home and away goal distributions. The comparison included the observed and posterior predictive mean goals and cumulative distribution plots displaying 100 posterior predictive samples. These checks examined whether the model could reproduce the general scoring patterns present in the data.

4.4.3 Expected goals and credible intervals

For each validation or test match, every retained posterior sample produced a different expected home and away goal rate:

$$\lambda_{H,i}^{(s)} = \exp\left(\beta_{0,H}^{(s)} + \mathbf{x}_i^T \boldsymbol{\beta}_H^{(s)}\right) \qquad \lambda_{A,i}^{(s)} = \exp\left(\beta_{0,A}^{(s)} + \mathbf{x}_i^T \boldsymbol{\beta}_A^{(s)}\right)$$

where $s = 1, \ldots, 4000$ represents a posterior draw. The posterior mean expected goals were calculated as:

$$\bar{\lambda}_{H,i} = \frac{1}{4000} \sum_{s=1}^{4000} \lambda_{H,i}^{(s)} \qquad \text{and} \qquad \bar{\lambda}_{A,i} = \frac{1}{4000} \sum_{s=1}^{4000} \lambda_{A,i}^{(s)}$$

The 2.5^{th} and 97.5^{th} percentiles of the posterior rate draws formed a 95% credible interval for each team's expected goal rate. These intervals represent uncertainty concerning the expected number of goals resulting from the uncertainty in the fitted coefficients. They are not prediction intervals for the actual number of goals scored, which would additionally contain match level Poisson variation.

4.4.4 Converting posterior draws into result probabilities

Home win, draw and away win probabilities were calculated separately for every posterior draw. If home and away goals are conditionally independent Poisson variables their difference follows a Skellam distribution. Defining the goal difference for posterior draws as:

$$D_i^{(s)} = Y_{H,i}^{(s)} - Y_{A,i}^{(s)}.$$

the three result probabilities were

$$P\left(D_i^{(s)} > 0\right), \qquad P\left(D_i^{(s)} = 0\right), \qquad P\left(D_i^{(s)} < 0\right).$$

The final probability for each result was obtained by averaging over all 4,000 posterior draws:

$$\bar{P}_i(k) = \frac{1}{4000} \sum_{s=1}^{4000} P_i^{(s)}(k), \qquad k \in \{H, D, A\}.$$

Unlike the score matrix calculation used for the standard Poisson model, Skellam calculations did not require goal totals to be truncated at ten. The predicted match result was the outcome with the largest posterior averaged probability.

The Bayesian model continued to assume conditional independence between home and away goals and retained the standard Poisson likelihood. Its principal difference from the standard Poisson regression was therefore the representation and propagation of coefficient uncertainty. The posterior averaged home win, draw and away win probabilities were subsequently used in the Monte Carlo season simulations.

4.5 Monte Carlo Season Simulation

Monte Carlo simulation was used to convert the match predictions into possible final league tables for the 2025/26 Premier League season. Instead of assuming that the most probable outcome would occur in every fixture, each match was simulated according to the uncertainty represented by the relevant model. Repeating this process produced a distribution of possible points totals, finishing positions and season outcomes.

Separate simulations were conducted for the selected machine learning model, the standard Poisson and the Bayesian Poisson model. Each model was used to simulate the complete set of 380 matches 10,000 times.

4.5.1 Machine learning simulation

The machine learning classifier directly produced home win, draw and away win probabilities for every test match. In each simulation the result of match i was sampled from a categorical distribution:

$$R_i^{(s)} \sim \text{Categorical}\left(p_{H,i}, p_{D,i}, p_{A,i}\right)$$

where $p_{H,i}$, $p_{D,i}$ and $p_{A,i}$ are the model's predicted probabilities of a home win, draw and away win. Therefore an outcome with a higher predicted probability was more likely to be selected but less likely outcomes remained possible. The simulated result was converted into league points using the standard allocation:

$$\begin{cases} 3, & \text{for a win,} \\ 1, & \text{for a draw,} \\ 0, & \text{for a loss.} \end{cases}$$

The played, won, draw, lost and points columns were updated after fixture. Because the machine learning model predicted match results rather than exact scores as goals scored and conceded were not available. Teams finishing on the same number simulated points were therefore separated using a randomly generated tie-breaker. This prevented alphabetical team order from determining their finishing positions.

4.5.2 Standard Poisson simulation

The standard Poisson simulation used expected home and away goal rates produced for each fixture. For the simulations, goals were independently generated as:

$$G_{H,i}^{(s)} \sim \text{Poisson}\left(\widehat{\lambda}_{H,i}\right) \qquad \text{and} \qquad G_{A,i}^{(s)} \sim \text{Poisson}\left(\widehat{\lambda}_{A,i}\right)$$

where $\widehat{\lambda}_{H,i}$ and $\widehat{\lambda}_{A,i}$ are the expected home and away goals from the fitted standard Poisson models.
The simulated score determined whether the match ended in a home win, draw or away win. Points were then awarded using the standard system. Because exact scores were generated each team's goals scored, goals conceded and goal difference was recorded as well. At the end of each simulated season, clubs were ranked by points, followed by goal difference then goals scored. This reproduced the principal ranking criteria used in the Premier League and provided a more informative tie-breaker procedure that was available for machine learning simulation.

4.5.3 Bayesian Poisson simulation

The Bayesian simulation incorporated both match level scoring variation and uncertainty in the fitted model parameters. At the beginning of each simulated season, one of the 4000 retained posterior draws was selected at random. The home and away expected goal rates from this posterior draw were then used for all 380 fixtures in that simulated season.
If q_s represents the posterior draw selected for simulation, the goals for match I were generated as:

$$G_{H.i}^{(s)} \sim \text{Poisson}\left(\lambda_{H,i}^{(q_s)}\right) \qquad \text{and} \qquad G_{A,i}^{(s)} \sim \text{Poisson}\left(\lambda_{A,i}^{(q_s)}\right)$$

Selecting one posterior draw for each complete season ensured that a consistent set of model coefficients was used across all fixtures within that simulation. A different posterior draw could be selected in another simulated season. Consequently, variation between simulations reflected uncertainty in the fitted coefficients as well as random variations in the goals scored during individual matches.
The simulated goals were converted into match results and league points. As with the standard Poisson simulations, teams were ranked on points, goal difference and goals scored.

4.5.4 Simulation summaries

For team t the expected points total was calculated by averaging its points across the $S = 10000$ simulated seasons:

$$\frac{1}{S}\sum_{s=1}^{S} P_t^{(s)}$$

where $P_t^{(s)}$ is the points total obtained by team t in simulation s.
The 5th and the 95th percentile of each team's simulated position were also recorded. The 5th represented a favourable finishing position while the 95th was an unfavourable one. These percentiles provided a more informative range than reporting only the best and worst position observed in the 10,000 simulations.
Average wins, draws and losses were calculated for all three models. The standard and Bayesian Poisson simulations additionally produced expected goals scored, goals conceded and goal difference because they simulated using the exact scorelines.
Title probability was estimated as the proportion of simulations a club finished first:

$$\frac{1}{S}\sum_{s=1}^{S} \mathbb{I}\left(r_t^{(s)} = 1\right).$$

Top four probability was then calculated as:

$$\frac{1}{S}\sum_{s=1}^{S} \mathbb{I}\left(r_t^{(s)} \leq 4\right),$$

while relegation probability was:

$$\frac{1}{S}\sum_{s=1}^{S} \mathbb{I}\left(r_t^{(s))} \geq 18\right)$$

The resulting proportions were then multiplied by 100 to convert into percentages.
A final predicted table was then constructed for each model. The machine learning table was ordered by expected points while the standard and Bayesian Poisson used expected points followed by expected goal difference then expected goals scored. The match probabilities and preseason predictors were not updated during a simulated season. Therefore, simulated results from earlier fixtures did not alter the probabilities or goal rates for later matches. Each simulation represented one possible realisation of the season under the preseason information and fitted model available before the test season.

4.6 Evaluation Measures

Model Performance was evaluated at three levels: match result prediction, goal prediction and final league table prediction. This was necessary because a model that predicts individual matches may not necessarily reproduce the final structure of the season. Similarly, two models can have the same classification accuracy while assigning substantially different probabilities to their predictions.
The validation season was used to compare candidate models and select the final machine learning approach. After model selection each final model was fitted using the combined training and validation dataset and evaluated on the separate 2025/26 test season. The test data were not used for feature construction, parameter estimation or model selection.

4.6.1 Match result evaluation

Match result performance was evaluated using accuracy, macro F1-score and log loss. Confusion matrices and class specific precision, recall and F1-scores were examined to identify differences in the prediction of home wins, draws and away wins. Accuracy measured the proportion of matches for which the predicted result was correct:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left(\widehat{y}_i = y_i\right)$$

where N is the number of matches, y_i is the actual result and $\widehat{y}_i$ is the predicted result. The indicator function $\mathbb{I}(\cdot)$ equals one when the prediction is correct and zero otherwise. Higher accuracy indicates stronger classification performance. Accuracy can be influenced by the unequal frequencies of home wins, draws and away wins. Macro F1-score was therefore included to give equal importance to all three result classes. For result class k, precision and recall were defined as:

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k} \qquad \text{and} \qquad \text{Recall}_k = \frac{TP_k}{TP_k + FN_k}$$

where TP_k, FP_k and FN_k represent the true positives, false positives and false negatives for class k. The class specific F1-score was:

$$F1_k = 2\left(\frac{\text{Precision}_k\,\text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}\right)$$

Consequently a model could not obtain a strong macro F1 score simply by predicting the most common result accurately. If a model made no predictions for a particular class, the undefined precision and F1 values for that class were treated as a zero.
Log loss evaluated at the complete probability distribution assigned to the three possible outcomes:

$$-\frac{1}{N}\sum_{i=1}^{N}\sum_{k\in\{H,D,A\}} z_{ik}\log(p_{ik})$$

where $z_{ik} = 1$ when class k is the actual result of the match i and zero otherwise, while p_{ik} is the probability assigned to that result. Lower log loss indicates better probabilistic prediction. The measure gives a particularly large penalty when a model assigns a very small probability to an outcome that subsequently occurs. Accuracy and macro F1 assessed the most likely predicted result, whereas log loss assessed the quality and

confidence of the full probability distribution. These measures were therefore interpreted together rather than treating any single value as a complete description of model performance.

4.6.2 Goal prediction evaluation

Goal prediction measures were calculated only for the standard and Bayesian Poisson models. Mean absolute error was used to compare predicted expected goals with the observed goal totals:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \widehat{\lambda}_i\right|$$

where y_i is the observed number of goals and $\widehat{\lambda}_i$ is the predicted expected goal rate. Separate MAE values were calculated for home goals and away goals.
Total goal MAE was also calculated by comparing the observed total goals in each match with the sum of the predicted home and away goal rates:

$$\text{Total-goal MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|(y_{H,i} + y_{A,i}) - (\hat{\lambda}_{H,i} + \hat{\lambda}_{A,i})\right|$$

Lower MAE values indicate that the expected goal predictions were on average closer to the observed goal totals. Mean Poisson deviance was used as an additional measure because it evaluates count predictions under the Poisson distribution:

$$D = \frac{2}{N}\sum_{i=1}^{N}\left[y_i \log\left(\frac{y_i}{\hat{\lambda}_i}\right) - (y_i - \hat{\lambda}_i)\right]$$

When $y_i = 0$ the term involving the logarithm was defined as zero. Predicted goal rates were restricted to a minimum of 10^{-10} before calculating the deviance to prevent numerical problems caused by taking the logarithm of zero. Separate deviance values were calculated for home and away goals. Lower Poisson deviance indicated a better fit between the predicted scoring rates and the observed in counts. The Poisson goal predictions were also compared with a mean goal baseline. This baseline predicted every match using the average home and away goals observed in the relevant training data.

4.6.3 League table evaluation

The simulated league tables were compared with the actual 2025/26 final table using position. MAE, points MAE and Spearman rank correlation. The measures were calculated for the machine learning, standard Poisson and Bayesian Poisson simulations.
Position MAE measured the average absolute difference between each club's predicted and actual finishing positions:

$$\text{Position MAE} = \frac{1}{20}\sum_{t=1}^{20}|\widehat{r}_t - r_t|$$

where $\widehat{r}_t$ and r_t are the predicted and actual positions of team t. Lower position MAE indicates a more accurate predicted league order.

Points MAE measures the average difference between expected and actual points:

$$\text{Points MAE} = \frac{1}{20}\sum_{t=1}^{20}\left|\widehat{P}_t - P_t\right|$$

where $\widehat{P}_t$ is the team's average points across the 10,000 simulations and P_t is its actual points for the season. Spearman rank correlation measured the similarity between the complete predicted and actual league rankings:

$$\rho = 1 - \frac{6\sum_{t=1}^{20} d_t^2}{20(20^2-1)}$$

where d_t is the difference between the predicted and actual rank t. A value close to one indicates that the two tables have similar ordering, while a value close to zero indicated little relationship between them.
Several season outcome comparisons were also reported. The predicted champion was defined as the club with the highest simulated title probability with expected points used only as a secondary tie-breaker if required. Champion prediction was recorded as correct or incorrect. The number of predicted top four teams appearing in the actual top four and the relegation overlap were also measured. Together these measures allowed the model to be compared in terms of individual match predictions, expected goals, probability quality and their ability to reproduce the overall 2025/26 Premier League season.

5 Results

This chapter presents the validation and final test results for the baseline, machine learning, standard and Bayesian Poisson models. It explains the selection of the final machine learning model using the 2024/25 validation season. The selected models are then evaluated on the separate 2025/26 test season at match result, goal prediction and season level. Bayesian sampling diagnostics and posterior predictive checks are also presented before the simulated league tables and season outcome probabilities are compared.

5.1 Validation Results and Model Selection

The models were initially evaluated on the 380 matches from the 2024/25 validation season. This season contained 155 home wins, 93 draws and 132 away wins. Accuracy, macro F1-score and log loss were examined together, but log loss was the primary measure used to select the final machine learning model because complete outcome probabilities were required for the season simulations.

Table 5.1 presents the validation results. The majority class baseline achieved an accuracy of 0.408 because it predicted a home win for every match. Its macro F1-score was only 0.193 showing its failure to identify either draws or away wins and it had a log loss of 1.082.

Model	**Accuracy**	**Macro F1-score**	**Log loss**
Majority-class baseline	0.408	0.193	1.082
Logistic regression	0.532	0.400	1.004
Gradient boosting	0.479	0.391	1.034
Standard Poisson	0.526	0.395	0.992
Bayesian Poisson	0.526	0.395	1.002

Table 5.1: Match-result performance on the 2024/25 validation season

Logistic regression had the best performance of the two machine learning models with a validation accuracy of 53.2% while gradient boosting had 47.9%. Logistic regression also had the lower loss of 1.004 while gradient boosting achieved 1.034 and a slightly higher macro F1-score. It therefore outperformed gradient boosting in all measures.

Consequently logistic regression was selected as the final machine learning model. This result differed from Baboota and Kaur (2019) [1] in which gradient boosting was the strongest machine learning approach. However the present dissertation used different season, predictors and a preseason only prediction setting. The result demonstrates why model selection was performed using chronologically separate validation data rather than picking the strongest algorithm. The standard and Bayesian models were not included in the model selection as they represented a separate modelling approach compared to the machine learning options and were retained for the final comparison.

Draw prediction still remained difficult during validation. Logistic regression did not select a draw as the most likely result for any validation match while gradient boosting predicted 29 draws but only five of them were correctly identified which contributed to the lower macro F1-score despite stronger performance for the home and away wins.

5.2 Final Match Result Performance

Following validation logistic regression, standard Poisson regression and Bayesian Poisson regression were refitted using the combined training and validation dataset of 5,700 matches. The final models were then evaluated on the hold out 2025/26 season, which contained 380 matches.

Model	Test accuracy	Test macro F1	Test log loss
Majority-class baseline	0.426	0.199	1.082
Logistic regression	0.468	0.351	1.039
Standard Poisson	0.474	0.352	1.052
Bayesian Poisson	0.474	0.354	1.038

Table 5.2: Final match-result performance on the 2025/26 test season

Table 5.2 presents the accuracy, macro F1-score and log loss obtained by each model. All three fitted models outperformed the majority class baseline on every measure. The baseline predicted every match as a home win and correctly classified 162 of the 380 matches giving it an accuracy of 0.426. Its macro F1-score was 0.199 due to not identifying any away wins or draws.

The standard and Bayesian Poisson models achieved the highest test accuracy at 0.474 by classifying 180 matches correctly. Logistic regression achieved 0.468 at 178 matches correctly classified. The difference between the three models was very small with only two games separating logistic from the two Poisson models. The Bayesian Poisson had the highest macro F1-score at 0.354 followed closely by the standard Poisson at 0.352 and logistic regression at 0.351. The similarity of the scores indicates that the models produced comparable class level predictions.

The Bayesian model also achieved the lowest test log loss at 1.038. Logistic regression produced an almost identical value of 1.039 while standard Poisson had the highest at 1.052. Therefore although the two Poisson models had the same classification accuracy, the Bayesian model assigned slightly better probabilities to the observed outcomes. The standard Poisson model had produced the lowest loss during validation but this did not continue into the test season.

Model	Home win	Draw	Away win
Logistic regression	0.59	0.00	0.46
Standard Poisson	0.60	0.00	0.45
Bayesian Poisson	0.60	0.00	0.46

Table 5.3: Class level F1-scores on the 2025/26 test season

The test season had 162 home wins, 104 draws and 114 away wins. Home wins were predicted most successfully by all three models, away wins were slightly weaker but all three also had similar F1-score.

The main weakness was draw classification, none of the three final models selected a draw as the most likely outcome for any test match. Consequently all 104 draws were classified as either a home or away win, giving the draw class precision, recall and F1-score of zero. This does not mean that there was a zero probability

of a draw but that the draw probability was never larger than home or away win for each match.
Overall no model was clearly superior across all measures. The standard and Bayesian Poisson models achieved the highest accuracy while the Bayesian model obtained the strongest macro F1-score and lowest log loss. Logistic regression remained competitive and produced a log loss only 0.001 higher than the Bayesian model.

5.3 Goal Prediction Performance

Only the standard and Bayesian Poisson models produced predictions for the number of home and away goals. Their final goal prediction performance was evaluated using mean absolute error and mean Poisson deviance. Lower values indicate better performance for both measures. The Bayesian results were calculated using the posterior mean expected goal rate for each match.

Model	Home goal MAE	Away goal MAE	Total goal MAE	Home goal deviance	Away goal deviance
Standard Poisson	0.949	0.851	1.228	1.070	1.091
Bayesian Poisson	0.943	0.850	1.235	1.049	1.085

Table 5.4: Goal prediction performance on the 2025/26 test season

The Bayesian model produced the lower home goal MAE although the difference was small with it being approximately 0.943 while the standard Poisson model had 0.949. The away goal MAE had an even smaller difference of 0.001. The standard Poisson did perform slightly better when predicting the total goals in a match with an MAE of 1.228 while Bayesian model had 1.235.
The Bayesian model did achieve a lower mean Poisson deviance for both goal outcomes. Its home goal deviance was 1.049 compared with 1.070 for the standard model, its away goal deviance it had 1.085 compared with 1.091 for the standard model. These results indicate that the Bayesian posterior mean goal rates were more consistent with the observed goal counts under the Poisson loss function.
Both models produced a higher MAE for home goals compared to away goals MAE, however their away goal deviances were slightly higher than their home goals deviances. This difference occurs due to the mean absolute error measures the average size of the prediction error whereas the Poisson deviance evaluates the predicted rates relative to the probability structure assumed by the Poisson model. Overall the differences between the two approaches were limited. The Bayesian model performed marginally better for home and away goals and the deviance measure while the standard Poisson produced lower total goal MAE. The Bayesian representation of coefficient uncertainty did not create a substantial improvement in point prediction accuracy.

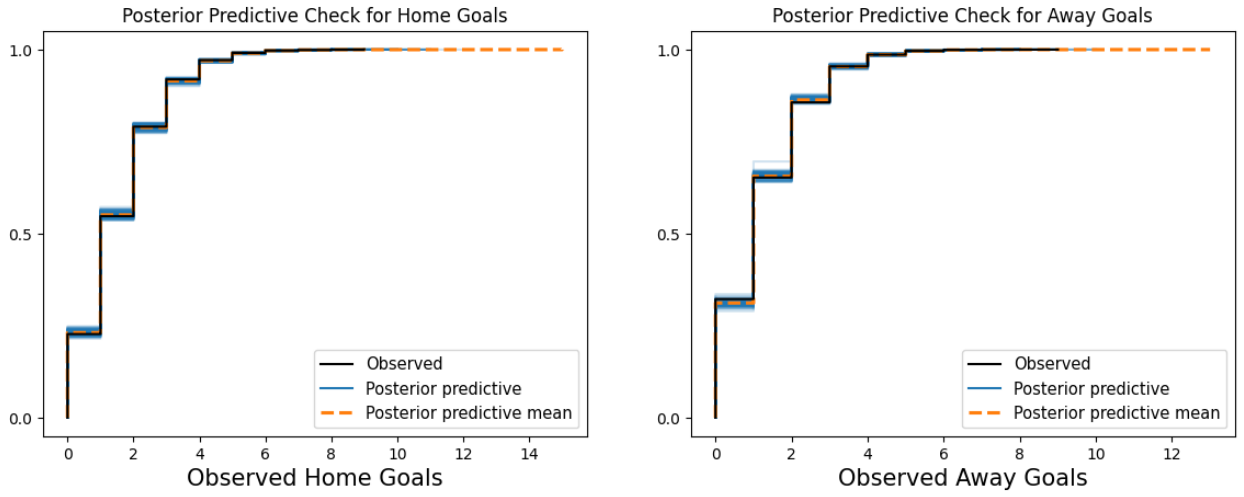
5.4 Bayesian Diagnostics and Uncertainty

The reliability of the Bayesian results depends on whether the posterior sampler converged and explored the posterior distributions effectively. Table 5.5 summarises the diagnostics for the validation stage and final refitted model.

Diagnostics	Validation stage model	Final refitted model
Maximum $\hat{R}$	1.005	1.004
Minimum bulk effective sample size	3,938.135	4,098.327
Minimum tail effective sample size	2,702.130	2,519.192
Divergent transitions	0	0

Table 5.5: Bayesian sampling diagnostics

The maximum $\hat{R}$ values were close to one for both fitted models. The final refitted model had a maximum value of 1.004 indicating that the four sampling chains produced very similar posterior distributions. No parameter had an $\hat{R}$ value above 1.01. The minimum bulk and tail effective sample sizes were also very high. For the final fitted model the minimum bulk effective sample size was approximately 4,098 and the minimum tail effective sample size was approximately 2,519. These values indicate that the 4,000 retained posterior draws contained substantial information including within the tails of the distributions.
No divergent transitions occurred during either sampling procedure. Taken together the $\hat{R}$, effective sample size and divergence results provided no evidence of important sampling or convergence problems. Posterior predictive checks were conducted using the validation stage Bayesian model fitted to training data. The observed mean and the mean across the posterior predictive home goals samples had the same value of 1.559, the same was for the away goals at 1.234.



Home goals Away goals

Figure 5.1: Posterior predictive checks for home and away goals.

The cumulative distribution plots compared the observed goal distributions with 100 posterior predictive samples. For both home and away goals the observed cumulative distribution was closely followed by the simulated distributions, particularly across the low and moderate goal counts containing most matches. Therefore, the model reproduced the overall location and shape of the training goal distributions reasonably closely.

The Bayesian model additionally produced 4,000 expected home goal and away goal rate draws for every match in the 2025/26 test season. The posterior means formed the point predictions evaluated in Section 5.3, while the 2.5^{th} and the 97.5^{th} percentiles formed the 95% credible interval.

For example, the opening match between Liverpool and Bournemouth had a posterior mean expected home goal rate of 2.053 with a 95% credible interval from 1.929 to 2.182. Bournemouth's expected away goal rate was 0.970 with a corresponding interval from 0.898 to 1.049.

These credible intervals describe uncertainty in each team's expected scoring rate resulting from uncertainty in the fitted coefficients. They should not be interpreted as intervals which the actual goal totals must fall within. Actual goals contain additional match level Poisson variation and can therefore lie outside the credible intervals for the expected rate. Overall the diagnostic results support the use of the posterior draws in the subsequent predictions. The Bayesian model successfully represented uncertainty in its coefficients and expected goal rates, although Section 5.3 showed that this produced only a small improvement over the standard Poisson model in point prediction performance.

5.5 League Table and Season Simulation Results

Each final model was used to simulate the complete 2025/26 Premier League season 10,000 times. The simulations produced expected points, average finishing positions, position-percentile ranges and probabilities of winning the title, finishing in the top four and being relegated. The representative predicted tables were ordered by expected points. Expected goal difference and expected goals scored were used as additional ordering criteria for the two Poisson models. The predicted champion was identified separately as the team with the highest title probability.

The logistic regression simulation placed Liverpool first with 74.06 expected points, followed by Manchester City with 73.48 points, Arsenal with 72.96 points then Chelsea with 63.47 points. Liverpool had the highest probability of winning the title with 33.66% compared to Manchester City with 30.35%, therefore the model selected Liverpool as the predicted champion.

The standard Poisson simulation also had Liverpool winning with 72.13 expected points but followed by Arsenal with 68.17 points, Manchester City with 67.13 points and Chelsea with 63.28 points. Liverpool again had the highest title probability with 41.00% which was substantially higher than Arsenal at 21.35%. The standard Poisson model consequently provided the strongest preference for Liverpool among the three approaches.

The Bayesian simulation produced a closer contest between Manchester City, Liverpool and Arsenal. Manchester City came first with 73.05 expected points with Liverpool very closely behind with 72.85 points and Arsenal with 72.23 points. Chelsea was expected fourth with 63.29 points. Manchester City also had the highest title probability at 32.82% compared to Liverpool at 31.17% and Arsenal at 27.18%. Manchester City was therefore selected as the Bayesian model's predicted champion.

The Bayesian simulations also demonstrated substantial uncertainty in the final positions. Manchester City, Liverpool and Arsenal each had a $5^{\text{th}} - 95^{\text{th}}$ percentile finishing position range of first to sixth. At the lower end of the table Sunderland and Leeds had ranges from thirteenth to twentieth while Burnley's range was fifteenth to twentieth. These intervals showed the average result across all simulations and that considerably different final positions were possible.

All three models predicted Sunderland, Leeds and Burnley to occupy the relegation positions. Under lo-

gistic regression, their probabilities were 67.89%, 71.51% and 87.06% respectively. The standard Poisson had 77.26%, 78.84% and 81.37% while Bayesian probabilities were 62.65%, 65.61% and 83.02% respectively. Burnley was considered by all three models to perform the worst in the 2025/26 season.

Model	Position MAE	Points MAE	Spearman correlation
Logistic regression	3.90	9.30	0.657
Standard Poisson	3.70	10.16	0.633
Bayesian Poisson	3.70	9.16	0.675

Table 5.6: Final league table prediction performance

The predicted tables were compared with the actual 2025/26 league tables using position mean absolute error, points mean absolute error and Spearman rank correlation. Table 5.6 shows the differences.
The Bayesian and standard Poisson models achieved the lowest position MAE at 3.70 compared to logistic regression at 3.90. However the Bayesian model produced the lowest points MAE at 9.16 and the highest Spearman correlation at 0.675. This indicated that it provided the strongest overall reconstruction of the ordering and points totals in the final league table.

Actual and Predicted 2025/26 Season Outcomes

Model	Champion	Top Four	Relegated
Actual 2025/26	Arsenal	Arsenal Man City Man United Aston Villa	West Ham Burnley Wolves
Logistic Regression	Liverpool	Liverpool Man City Arsenal Chelsea	Sunderland Leeds Burnley
Standard Poisson	Liverpool	Liverpool Arsenal Man City Chelsea	Sunderland Leeds Burnley
Bayesian Poisson	Man City	Man City Liverpool Arsenal Chelsea	Sunderland Leeds Burnley

Figure 5.2: Actual and Predicted 2025/26 Premier League season outcomes.

None of the models correctly predicted Arsenal as the League champion as shown in Figure 5.2. Logistic regression and standard Poisson had Liverpool as the expected winner while the Bayesian model had selected Manchester City. All three models had however predicted the same top four teams as Liverpool, Manchester City, Arsenal and Chelsea although there was some ordering difference. Out of them only Arsenal and Manchester City ended up finishing in the top four in the 2025/26 Premier League season. The models also had the same group of teams that were predicted to be relegated as shown in Figure 5.2. Only Burnley was actually relegated.

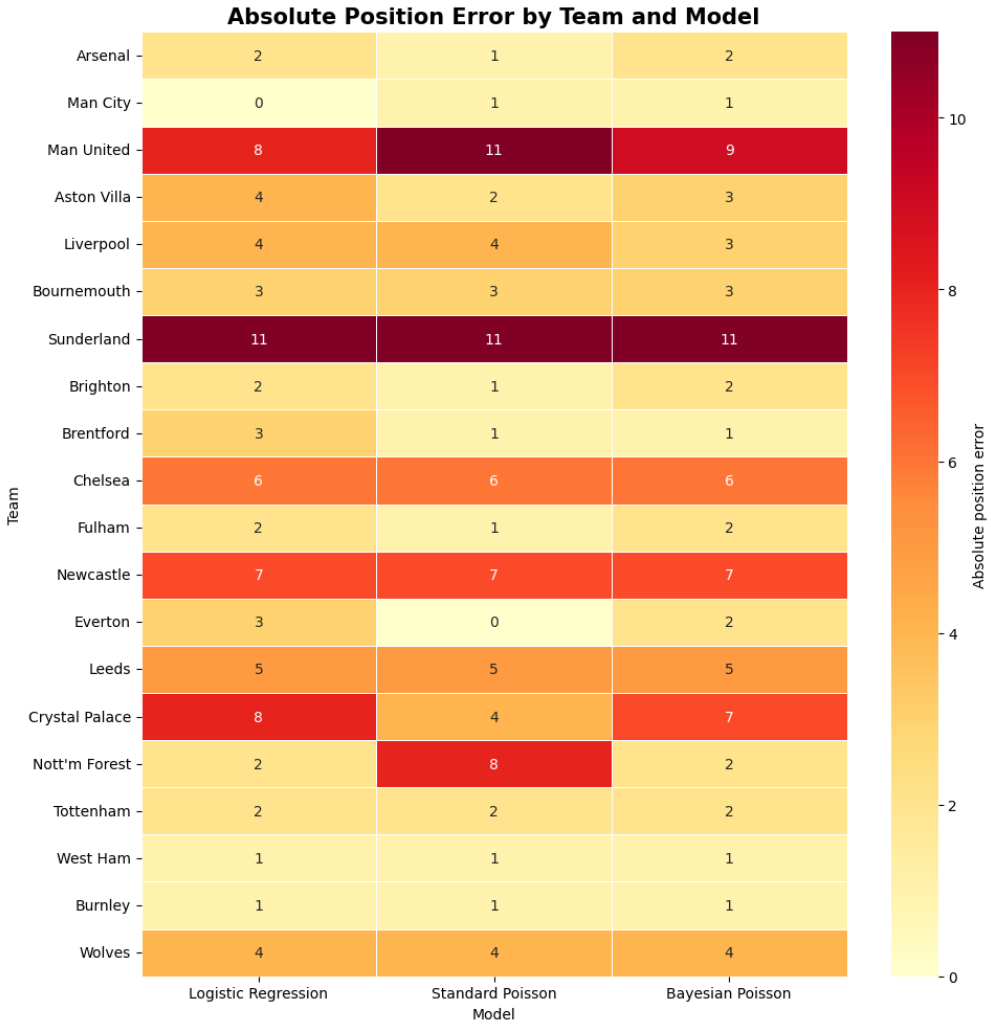


Figure 5.3: Position error between Actual and Predicted 2025/26 Premier League table

As seen in Figure 5.3 Sunderland represented the largest prediction error because it was expected to finish eighteenth but finished seventh. Manchester United was also underestimated while Chelsea and Newcastle were overestimated finishing considerably lower than expected.

Overall the Bayesian Poisson model produced the strongest league table evaluation. Nevertheless the errors with teams such as Arsenal, Sunderland and Chelsea demonstrate the difficulty of forecasting a complete season using only preseason information. Performances that differed substantially from the previous season could not be incorporated because the predictors remained fixed throughout the test season.

6 Discussion and Limitations

This chapter interprets the findings in Chapter 5 in relation to the research aim, questions and previous literature. The discussion considers the relative performance of logistic regression, standard Poisson and Bayesian Poisson regression at match, goal and season levels. It also examines the additional information provided by the Bayesian approach, explains the difference between the predicted and actual league tables and identifies the principal limitations of the study.

6.1 Overview of the Main Findings

The results demonstrate that no single modelling approach was clearly superior across every evaluation measure. The standard Bayesian Poisson models achieved the highest match result accuracy, while the Bayesian Poisson models achieved the highest macro F1-score and log loss. At the season level, the Bayesian model also achieved the lowest point MAE and highest Spearman correlation while matching the standard Poisson model's position MAE. It was therefore the strongest overall approach although the difference between the models were generally small.

The findings also show that preseason team strength information could identify broad differences between teams but could not fully represent the changes occurring during the predicted season. All three models produced reasonable probabilities for home and away wins but none selected a draw as the most likely result for any test match. They also made similar errors when predicting the champion, top four teams and relegated clubs. These results demonstrate the difficulty of predicting a complete season when the model inputs remain fixed from the beginning of the season.

The following sections examine these findings in greater details. Section 6.2 compares the advantages and disadvantages of the three modelling approaches, Section 6.3 discusses the value of the Bayesian uncertainty estimates, Section 6.4 interprets the season simulation results and Section 6.5 considers the limitations of the data and methodology.

6.2 Comparison of the Modelling Approaches

The three models differed mainly in what they predicted and how they represented uncertainty. Logistic regression directly estimated home win, draw and away win probabilities, whereas the two Poisson models first predicted home and away goals. This allowed the Poisson models to produce expected scores and use goal difference and goals scored when constructing their simulated league tables.

Logistic regression was the simplest and least computationally demanding approach. It remained competitive on the test season and produced a log loss that was almost identical to that of the Bayesian model. However logistic regression could not produce goal predictions or exact scorelines, making it less informative for complete season analysis.

The standard Poisson model achieved slightly higher match result accuracy and produced the lowest total goal MAE. Its score based structure also made it more suitable for simulating league tables. However it assumed that home and away goals were conditionally independent and did not include the low scoring adjustment proposed by Dixon and Coles (1997) [3]. These assumptions may have limited its representation of some football scorelines.

The Bayesian Poisson model used the same predictors and likelihood as the standard model, which explains why their results were similar. Its main advantage was the representation of coefficient and expected goal uncertainty through posterior distributions. It also produced the strongest overall league table performance. Nevertheless the improvement in point predictions was small, showing the Bayesian estimation did not remove limitations caused by the shared predictors and Poisson assumptions. It also required substantially more computation than the other approaches.
Overall logistic regression offered simplicity and competitive match probabilities, while the standard Poisson provided more useful score based predictions with lower computational cost than the Bayesian approach. The Bayesian model produced the most complete analysis and the strongest overall results, but its greatest contributions was the additional representation of uncertainty rather than a larger improvement in predictive accuracy.

6.3 Bayesian Uncertainty and Model Diagnostics

The main additional information provided by the Bayesian model was a distribution of plausible expected goal rates rather than only a single estimate for each match. The posterior means provided the point predictions used for evaluation, while the 95% credible interval showed the uncertainty in those expected rates resulting from uncertainty in the fitted coefficients. This made it possible to assess how precisely the model estimated a team's scoring rate.
The convergence diagnostics supported the reliability of the posterior samples. The values of $\hat{R}$ were close to one, the effective sample sizes were high and no divergent transitions occurred. The posterior predictive checks also showed that the model reproduced the general distribution and average number of home and away goals in the training data. The findings indicate that the sampling procedure operated successfully although the do not guarantee accurate predictions on unseen matches.
The credible intervals should not be interpreted as ranges within which the actual goals must fall because the do no include all match level Poisson variation. Their value is instead that they communicate the degree of confidence in the expected goal estimates and allow parameter uncertainty to influence the season simulations.

6.4 Interpretation of the Season Simulation Results

None of the models correctly predicted Arsenal as champion, while all three produced the same top four and relegation selections. This similarity suggests that the shared preseason predictors influenced the season forecasts more strongly than the differences between the modelling approaches. Although the Bayesian model produced the strongest overall league table measures, it could not anticipate every important change in team performance.
Sunderland represented the largest error having been predicted to finish eighteenth by every model but actually finishing seventh. As a promoted club Sunderland entered the model with relatively weak preseason information and a lower expected strength than established Premier League teams. Their unexpectedly string season could not be recognised because predictors were not updated after the season began. The same limitation contributed to Manchester United being underestimated and Chelsea and Newcastle being overestimated.

The predicted tables should therefore be interpreted as probability based forecasts rather than exact predictions. Expected points and positions summarised the average results across 10,000 possible seasons, while the percentile ranges demonstrated that many alternative finishing positions remained possible. The simulation results were useful for identifying broad differences in team strength and measuring uncertainty.

6.5 Limitations and Future work

The main limitations of this study was its reliance on preseason information. The predictors were based primarily on previous season performance and remained fixed throughout the 2025/266 test season. Consequently the models could not respond to changes in form, transfers, injuries, managerial changes or tactical developments. Future work could update team strength measures after each round of matches using recent results, goals, shots and Elo ratings. This would allow the forecasts to respond when a team's performance differed from its preseason expectations.

Promoted clubs also had less recent Premier League information than established teams. The promoted team indicators and Elo adjustments provided a practical solution but they could not fully represent differences in quality between promoted clubs. Future models could incorporate Championship performance adjusted for differences in league strength, alongside squad changes and transfer information. This may provide a more accurate starting strength for newly promoted and returning teams.

Both Poisson models assumed that home and away goals were conditionally independent and followed standard Poisson distributions. They therefore did not account directly for dependence between the two teams' scores or unusual probabilities among low scoring outcomes. Future research could apply the low score adjustment proposed by Dixon and Coles (1997) [3], use bivariate Poisson distribution or allow team strengths to change dynamically as considered by Koopman and Lit (2015) [4]. These extensions may also help address the difficulty every model experienced when predicting draws.

The Bayesian model was feature based rather than hierarchical and therefore did not estimate partially pooled attacking and defensive strength for individual teams. A future hierarchical Bayesian model, similar to the approach of Baio and Blangiardo (2010) [5]could provide team specific parameters while sharing information across clubs. Further work could also examine the sensitivity of the results to different prior distributions and model specifications.

Finally, model selection and final evaluation each relied on a single football season. Although the chronological split prevented information leakage, the reported performance may have been influenced by season specific events. Rolling validation and testing across several seasons would provide stronger evidence about whether the relative performance of the models remained consistent. The season simulations could also be extended so that simulated results update team strengths and influence the probabilities of later matches. These improvements would create a more realistic representation of how team performance and uncertainty develop throughout a season.

7 Conclusion

This dissertation examined whether preseason information could be used to predict Premier League match results, goals and final league positions. It compared logistic regression, standard Poisson regression and Bayesian Poisson regression. The 2025/26 season was kept separate and used to test the final models.

The research objectives were achieved by creating 16 predictors from previous season performance, Elo ratings and promoted team information. Logistic regression was selected as the final machine learning model because it performed better than gradient boosting on the validation season. The same predictors were then used in the Standard and Bayesian Poisson models.

On the test season the standard and Bayesian Poisson models achieved an accuracy of 47.4% compared with 46.8% for logistic regression. The Bayesian model also achieved the best macro F1-score and log loss, although the differences was small. A main weakness was draw prediction as none of the models selected a draw as the most likely result for any match.

The two Poisson models produced similar goal predictions. However the Bayesian model also provided posterior distributions and credible intervals. These showed the uncertainty in the expected goal estimates which was not provided by the standard Poisson model.

The Bayesian model produced the strongest overall league table results. It achieved a position MAE of 3.70, a points MAE 9.16 and a Spearman correlation of 0.675. However none of the models correctly predicted Arsenal as champion. Each model also correctly predicted only two of the actual top four and one of the three relegated teams.

Overall preseason information was useful for identifying general differences in team strength. However it could not account for changes that happened after the season began. The Bayesian Poisson model was the strongest overall model but its main advantage was the extra information it provided about uncertainty rather than a large improvement in prediction accuracy.

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